

Habib University

Software Engineering (Spring 2026)

Project Title

Courtify

(A Web-Based Sports Arena Booking and Management System)

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PROJECT TYPE

Courtify is a **full-stack, web-based booking and management information system**.

The system is designed to support end-to-end interaction between users, court owners, and administrators by integrating a client-side interface, server-side application logic, and a centralized relational database for persistent data storage.

HIGH-LEVEL SCOPE

Project Description

Courtify is a full-stack web-responsive application designed to streamline the discovery, booking, and management of sports courts across Karachi.

The platform supports three core user roles:

1. **Players (General Users):** Can find nearby sports arenas and book available courts through an intuitive interface, optimized for desktop as well as Mobile devices.
2. **Court Owners:** Manage listings, availability, and view booking activity.
3. **Moderators/Admins (Platform Owners):** Oversee operations, perform analytics, and maintain the integrity of the system by managing users and courts.

Drawing inspiration from platforms like **Foodpanda** (for users) and **Airbnb** (for listings), *Courtify* adapts these models for the **local sports ecosystem** by offering a robust court booking and arena management platform that works seamlessly across **web** environments.

Core Features and User Roles

1. Player (General User):

- Register/Login using email & password (with hashed storage)

- Email verification upon signup to ensure valid signups
- Search and filter arenas
- View available time slots
- Book courts and view booking history

2. Court Owner:

- Secure login and dashboard access
- Add/update court availability
- View/manage upcoming and past bookings with analytics

3. Admin (Moderator):

- Add/remove courts and court owners
- Manage user accounts
- View platform-wide analytics (most active courts, flagged users, etc.)
- Moderate suspicious listings and bookings

OUT OF SCOPE

To ensure the feasibility of the project within a single academic semester, the following features are explicitly excluded from the scope of Courtify:

- Online payment processing, billing, or financial transactions
- Automated refunds, dispute handling, or payment reconciliation
- Native mobile applications for Android or iOS platforms
- Real-time GPS-based location tracking of users
- Integration with third-party payment gateways or external booking platforms
- AI-based recommendations, demand forecasting, or dynamic pricing
- Large-scale scalability, load balancing, or production-grade infrastructure optimization

Trello Board Link

[https://trello.com/invite/b/698837336d9f703166db4a76/ATTIa851e3d013ad873500b9c ae9bf5feebe28032CF2/courtify](https://trello.com/invite/b/698837336d9f703166db4a76/ATTIa851e3d013ad873500b9c%20ae9bf5feebe28032CF2/courtify)

Project Overview

Tech Stack:

Frontend: React.js

Backend: Node.js

Database: MySQL

Authentication: JWT and Google OAuth

Expected Deliverables

- Fully deployed **responsive website**
- Role-specific dashboards for players, court owners, and admins
- O-Auth user authentication from Supabase
- Email verification system to ensure valid signups
- Live demo using mock booking and courts data
- Real-time booking validation to prevent double booking

Architecture Diagram

